Critical metrics on connected sums of Einstein four-manifolds

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- Can replace the first by $\int |W|^2 dv$, where W is the Weyl tensor.

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By the C-G-B formula,

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($t = \infty$ corresponds to $g \mapsto \int R^2 dv$).

The Weyl functional

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- Its gradient is the *Bach tensor*:

$$B_{ij} = -4\big(\nabla^k \nabla^\ell W_{ikj\ell} + \frac{1}{2} R^{k\ell} W_{ikj\ell}\big).$$

Critical metrics (i.e., metrics for which $B_{ij} = 0$) are called *Bach-flat*.

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- It follows that any metric which is locally conformal to an Einstein metric is also Bach-flat.

Bach-flat metrics, cont.

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$$\Lambda_2 = \Lambda_2^+ \oplus \Lambda_2^-$$

into sub-bundles of self-dual/anti-self-dual forms.

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(X^4, g) is called *self-dual* (resp., *anti-self-dual*) if $W^- \equiv 0$ (resp., $W^+ \equiv 0$.)

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Examples. Locally conformally flat manifolds, $(\mathbb{CP}^2, g_{FS})$.

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- By the Hirzebruch signature formula,

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- All known examples of (compact) Bach-flat manifolds are either SD/ASD or Einstein.

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Lemma

(X^4, g) is B^t-flat $\Leftrightarrow$ it has CSC and

$$B_{ij} = 2tRE_{ij},$$

where $E_{ij} = R_{ij} - \frac{1}{4}Rg_{ij}$ is the trace-free Ricci tensor.

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The Catch: We cannot specify *a priori* the value of t.

An overview of the gluing procedure

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We want to prove the existence of a B^t-flat metric on $X = Y \sharp Z$.

Step 1: Blowing up

• Since $R_Y = R(g_Y) > 0$, the conformal laplacian $L = -\frac{1}{6}\Delta_Y + R_Y > 0$. Therefore, the Green's function G exists, with pole at y_0:

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• By the formulas for the Green's function above, g_N is scalar-flat:

$$R_N = G^{-3} L G = 0.$$

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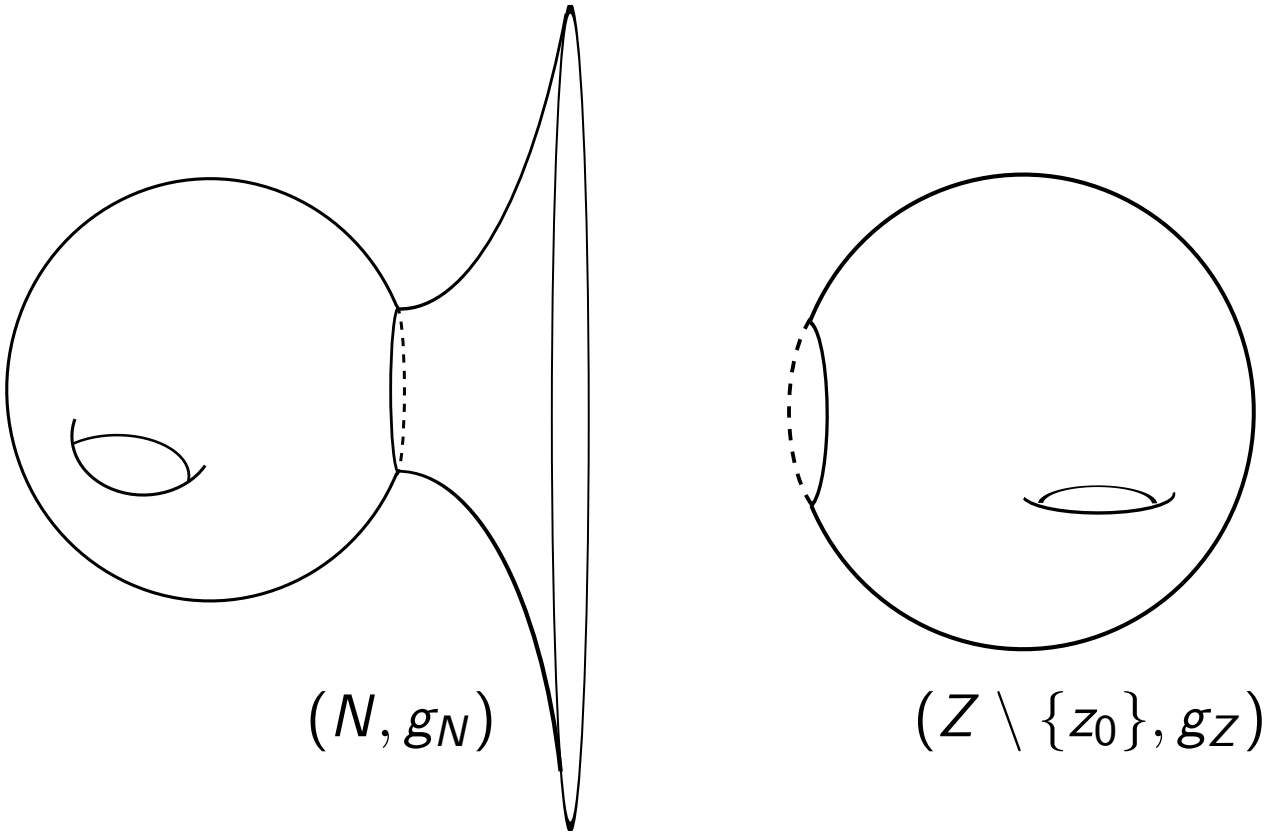
$$(g_N)_{ij} = \delta_{ij} + O(|x|^{-2}).$$

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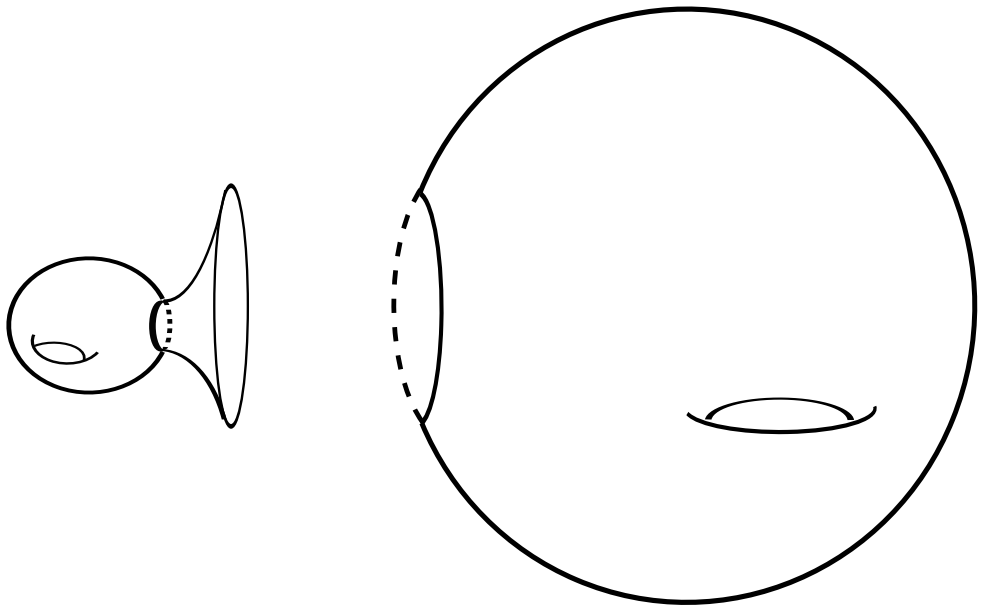
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(N, g_N) $(Z \setminus \{z_0\}, \tilde{g})$

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• On Z, choose g_Z-normal coordinates centered at z_0, and consider the annulus $A_Z = \{z \ : \ b < |z| < 2b\}$.

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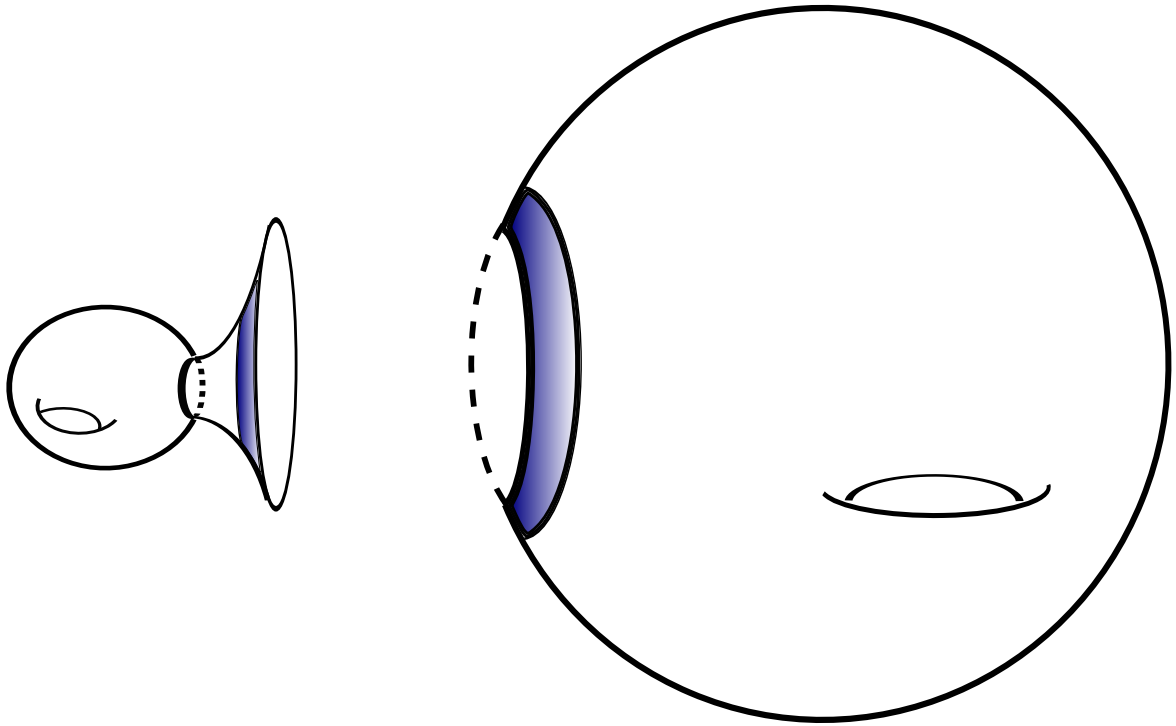
• On Z, choose g_Z-normal coordinates centered at z_0, and consider the annulus $A_Z = \{z \ : \ b < |z| < 2b\}$.

• On N, use the coordinates $\{x^i\}$ near infinity to define the annulus $A_N = \{x \ : \ a^{-1} < |x| < 2a^{-1} \}$.

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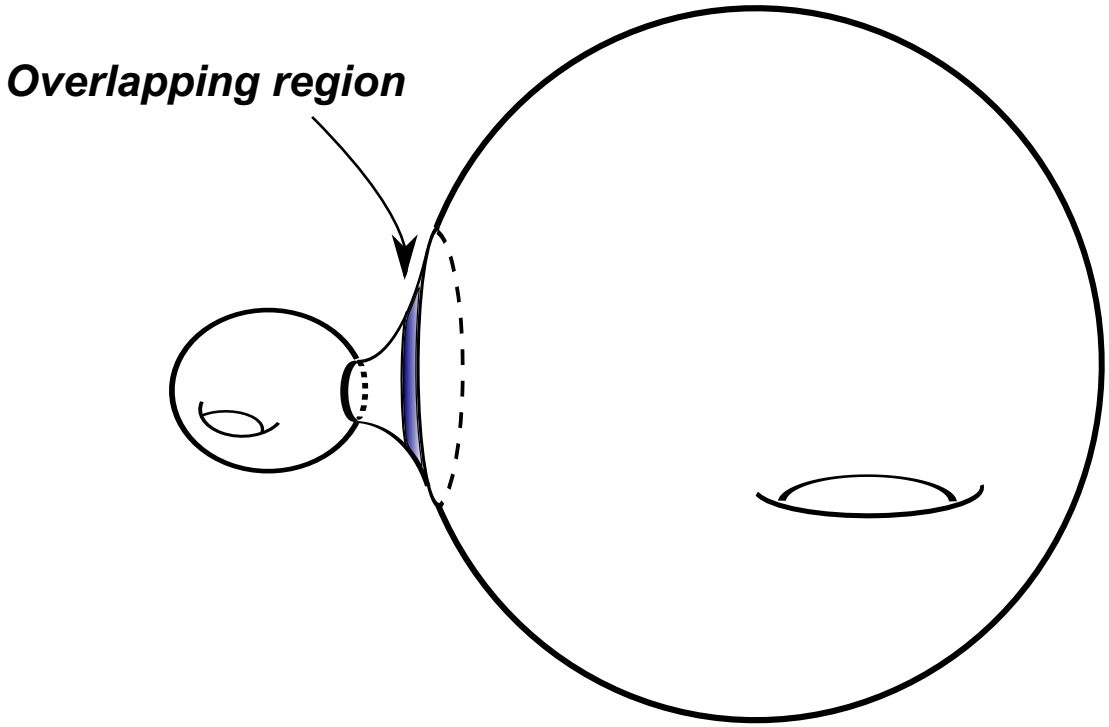
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- The tensor $B + tC$ of $g_{a,b}$ will satisfy

$$|(B + tC)(g_{a,b})| = O(a^4b^2) + O(a^6),$$

which unfortunately is too crude. (We need the metrics g_N and $\tilde{g}$ to 'match' to higher order.)

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• The operator Γ (for "gauge-fixing") is chosen so that the linearized operator is elliptic.

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(ii) The linearized operator S is elliptic.

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• In general, however, the cokernel will be non-trivial.

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- From now on we assume these symmetries for all metrics, functions, etc.

The linearized operator on $(Z \setminus \{z_0\}, g_Z)$

Theorem

(G- Viaclovsky, '12) For $t < 0$, the cokernel of $S = S_Z$ is

$$c \cdot g_Z$$

(which comes from scaling).

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- In fact, h decays quadratically.

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• For a, b sufficiently small we can then solve

$$P(\theta) = \lambda_1 \kappa_1 + \lambda_2 \kappa_2.$$

Solving modulo the cokernel(s), cont.

Key Proposition

Take $a = b$. Then

$$\lambda_1 = \mu a^4 + O(a^5),$$

where

$$\mu = \left\{ \frac{2}{3} W(y_0) \star W(z_0) + 4tR(z_0)\text{mass}(g_N) \right\} |S^3|.$$

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- Then an easy argument shows $\lambda_2 = 0$; i.e.,

$$P(\theta) = 0.$$

Statement of the Main Result

Theorem

(G - Viaclovsky) The following 4-manifolds admit a (toric-invariant) B^t-flat metric:

$$\mathbb{CP}^2 \sharp \overline{\mathbb{CP}}^2, \ \mathbb{CP}^2 \sharp 2\overline{\mathbb{CP}}^2, 2\sharp S^2 \times S^2.$$

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Topology of connected sum	Value(s) of t_0
$\mathbb{CP}^2 \sharp \overline{\mathbb{CP}}^2$	$-1/3$
$S^2 \times S^2 \sharp \overline{\mathbb{CP}}^2 = \mathbb{CP}^2 \sharp 2\overline{\mathbb{CP}}^2$	$-1/3$, $-.1892...$
$2\sharp S^2 \times S^2$	$-.3784...$

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- Trying to determine which happens is ongoing work.